

SF1811 Optimization

Lecture 7: Convexity

Chapters 8 and 15; see also Chapters 4 and 27

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Convexity

Main question:

Why does convexity make nonlinear optimization manageable?

Today:

- Convex sets.
- Extreme points.
- Convex functions.
- Convex optimization problems.
- Local and global minimizers.
- First- and second-order characterizations.

Convex sets

Definition: Convex set

A set $C \subseteq \mathbb{R}^n$ is convex if

$$x, y \in C, \quad t \in [0,1] \quad \Longrightarrow \quad (1-t)x + ty \in C.$$

The point

$$(1-t)x + ty = x + t(y-x)$$

traces the line segment from x to y as t varies from 0 to 1.

Convex combinations

A point of the form

$$(1-t)x + ty, \qquad t \in [0,1],$$

is a convex combination of x and y.

More generally,

$$\sum_{i=1}^{k} t_i x_i$$

is a convex combination of $x_1, \ldots, x_k$ if

$$t_i \geq 0, \qquad \sum_{i=1}^{k} t_i = 1.$$

If C is convex and $x_1, \ldots, x_k \in C$, then every convex combination of these points belongs to C.

A convex set contains all mixtures of its points.

The feasible region of a linear program is convex

Consider the feasible region

$$F = \{x \in \mathbb{R}^n : Ax = b,\ x \geq 0\}.$$

Let $x, y \in F$ and $t \in [0, 1]$. Then

$$\begin{aligned} A\big((1-t)x + ty\big) &= (1-t)Ax + tAy \\ &= (1-t)b + tb \\ &= b. \end{aligned}$$

Furthermore,

$$(1-t)x + ty \geq 0.$$

Hence,

$$(1-t)x + ty \in F.$$

Every linear-programming feasible region is convex.

Extreme points

Definition: Extreme point

Let C be convex. A point $x \in C$ is an extreme point of C if

$$x = (1-t)y + tz, \qquad y, z \in C, \qquad t \in (0,1),$$

implies

$$y = z = x.$$

Thus, an extreme point cannot be written as a proper convex combination of two different points in C.

For a linear program in standard form:

basic feasible solutions $\iff$ extreme points.

Why is a basic feasible solution an extreme point?

Let x be a basic feasible solution:

$$x_N = 0, \qquad A_B x_B = b, \qquad x_B \geq 0.$$

Suppose that

$$x = (1-t)y + tz, \qquad y, z \in F, \qquad t \in (0,1).$$

Since $y, z \geq 0$ and $x_N = 0$,

$$y_N = z_N = 0.$$

Therefore,

$$A_B y_B = b = A_B z_B.$$

Since A_B is nonsingular,

$$y_B = z_B = x_B.$$

Hence $y = z = x$, so x is an extreme point.

Convex functions

Definition: Convex function

Let $C \subseteq \mathbb{R}^n$ be convex. A function $f : C \to \mathbb{R}$ is convex on C if

$$f\big((1-t)x + ty\big) \leq (1-t)f(x) + tf(y)$$

for all $x, y \in C$ and $t \in [0, 1]$.

The graph of a convex function lies below every secant.

Examples of convex functions

An affine function

$$f(x) = c^{\mathsf{T}}x + c_0$$

satisfies

$$f\big((1-t)x + ty\big) = (1-t)f(x) + tf(y).$$

Hence, every affine function is both convex and concave.

The function

$$f(x) = x^2$$

is convex on $\mathbb{R}$, while

$$f(x) = -x^2$$

is concave on $\mathbb{R}$.

Sublevel sets of convex functions

Let $g : C \to \mathbb{R}$ be convex and let

$$F = \{x \in C : g(x) \leq 0\}.$$

If $x, y \in F$ and $t \in [0, 1]$, then

$$\begin{aligned} g\big((1-t)x + ty\big) &\leq (1-t)g(x) + tg(y) \\ &\leq 0. \end{aligned}$$

Thus,

$$(1-t)x + ty \in F.$$

Proposition

Every sublevel set of a convex function is convex.

Convex inequalities generate convex feasible regions.

Convex optimization problems

Consider

$$(P) \qquad \begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & x \in F. \end{array}$$

The problem is a convex optimization problem if

- the feasible region F is convex;
- the objective function f is convex on F.

A common form is

$$\begin{array}{lll} \text{minimize} & f(x) & \\ \text{subject to} & h_i(x) = 0, & i \in \mathcal{E}, \\ & g_i(x) \leq 0, & i \in \mathcal{I}, \end{array}$$

where f and the functions g_i are convex and the functions h_i are affine.

Linear programming is a special case of convex optimization.

Local and global minimizers

A feasible point $\hat{x} \in F$ is a local minimizer if there is an $\epsilon > 0$ such that

$$f(x) \geq f(\hat{x})$$

for every $x \in F$ satisfying $\|x - \hat{x}\| < \epsilon$.

A feasible point $\hat{x} \in F$ is a global minimizer if

$$f(x) \geq f(\hat{x})$$

for every $x \in F$.

For a convex problem, local means global

Theorem

Every local minimizer of a convex optimization problem is a global minimizer.

Proof

Suppose that $\hat{x}$ is a local minimizer but not a global minimizer. Then there is a $y \in F$ such that

$$f(y) < f(\hat{x}).$$

For $t \in (0, 1)$, let $x(t) = (1 - t)\hat{x} + ty$.
Since F is convex, $x(t) \in F$. Since f is convex,

$$f(x(t)) \leq (1 - t)f(\hat{x}) + tf(y) < f(\hat{x}).$$

For positive t sufficiently close to zero, $x(t)$ lies in any prescribed neighborhood of $\hat{x}$. This contradicts local optimality.

Why convexity matters

For a general nonlinear optimization problem:

- there may be many local minimizers;
- a local method may stop at a point that is not globally optimal;
- local information may not reveal the global structure.

For a convex optimization problem:

- every local minimizer is global;
- a point with a strictly smaller value can be approached along a feasible line segment;
- local optimality conditions become global optimality certificates.

In convex optimization, local information has global consequences.

First-order characterization of convexity

Let $C \subseteq \mathbb{R}^n$ be convex and let $f : C \to \mathbb{R}$ be continuously differentiable.

Theorem

The function f is convex on C if and only if

$$f(y) \geq f(x) + \nabla f(x)(y - x)$$

for all $x, y \in C$.

The function

$$f(x) + \nabla f(x)(y - x)$$

is the linearization of f at x, evaluated at y.

A differentiable convex function lies above every tangent plane.

Proof idea for the first-order characterization

Fix $x, y \in C$ and define the one-dimensional function

$$\varphi(t) = f\big(x + t(y - x)\big), \qquad t \in [0, 1].$$

Convexity gives, for $t \in (0, 1]$,

$$\varphi(t) \leq (1 - t)\varphi(0) + t\varphi(1).$$

Hence,

$$\frac{\varphi(t) - \varphi(0)}{t} \leq \varphi(1) - \varphi(0).$$

For positive t sufficiently close to zero, the left-hand side approaches

$$\varphi'(0) = \nabla f(x)(y - x).$$

Therefore,

$$f(y) \geq f(x) + \nabla f(x)(y - x).$$

Second-order characterization of convexity

Let $C \subseteq \mathbb{R}^n$ be open and convex, and let $f : C \to \mathbb{R}$ be twice continuously differentiable. Let $F(x)$ denote the Hessian of f at x.

Theorem

The function f is convex on C if and only if

$$d^{\mathsf{T}} F(x) d \geq 0$$

for all $x \in C$ and $d \in \mathbb{R}^n$.

Equivalently,

$$F(x) \succeq 0, \qquad x \in C.$$

A symmetric matrix H is

- positive semidefinite if $d^{\mathsf{T}} H d \geq 0$ for every d;
- positive definite if $d^{\mathsf{T}} H d > 0$ for every $d \neq 0$.

Quadratic functions

Consider

$$f(x) = \frac{1}{2} x^{\mathsf{T}} H x + c^{\mathsf{T}} x + c_0,$$

where H is symmetric.

Then

$$\nabla f(x) = x^{\mathsf{T}} H + c^{\mathsf{T}}$$

and

$$F(x) = H.$$

Therefore,

$$f \text{ is convex} \quad \iff \quad H \succeq 0.$$

If

$$H \succ 0,$$

then f is strictly convex and can have at most one minimizer.

For a quadratic function, convexity is determined entirely by the matrix H.

Main ideas

After this lecture, you should be able to:

- define a convex set and a convex function;
- show that the feasible region of a linear program is convex;
- define an extreme point and relate extreme points to basic feasible solutions;
- recognize a convex optimization problem;
- explain why every local minimizer of a convex problem is global;
- use first- and second-order characterizations of convexity;
- characterize convexity of a quadratic function in terms of its Hessian.

Convexity turns local structure into global information.